

Lena Collienue

Curriculum Vitae

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Research

Lecturer, *University of Otago*.

Developing deep learning methods for phylogenetics.

Postdoctoral Research, *Fred Hutchinson Cancer Center*,

(Advisor: Frederick Matsen IV, PhD).

Developing novel methods for phylogenetic inference using machine learning.

Postdoctoral Research, *University of Canterbury*,

(Advisor: Alex Gavryushkin, PhD).

Extending subtree prune and regraft operations to ranked phylogenetic trees and investigating thereby introduced distance measures for phylogenetic time trees.

PhD Research, *University of Otago*,

(Advisors: Alex Gavryushkin, PhD, and David Bryant, PhD).

Introducing and analysing spaces of phylogenetic time trees based on nearest neighbour interchange tree rearrangements.

M.Sc. Research, *University of Greifswald*,

(Advisors: Mareike Fischer, PhD, and Alex Gavryushkin, PhD).

Establishing properties of ranked nearest neighbour interchange moves between ranked phylogenetic trees

Education

2018–2021 **Doctor of Philosophy**, *Computer Science*, University of Otago (NZ).

2016–2018 **Master of Science**, *Biomathematics*, University of Greifswald (GER).

2012–2015 **Bachelor of Science**, *Biomathematics*, University of Greifswald (GER).

Work Experience

2026–present **Lecturer**, *School of Computing*, University of Otago (NZ).

2023–2026 **Postdoctoral Research Fellow**, *Matsen group*, Fred Hutch Cancer Center (US).

2022–2023 **Postdoctoral Research Fellow**, *BioDS lab*, *School of Mathematics and Statistics*, University of Canterbury (NZ).

Teaching Experience

2023–2024 **Facilitator**, *Girls Who Code*, Fred Hutch Cancer Center (Seattle, WA, USA).

2022 **Lecturer**, *STAT211: Random Processes*, University of Canterbury (NZ).

- 2019 **Tutor**, *COSC341: Theory of Computing*, University of Otago (NZ).
 2015–2016 **Summer Research Project**, *University of Auckland (NZ)*.

■ Scholarships and Awards

- 2022 *Hatherton Award* (Royal Society of New Zealand)
 2021 *Exceptional PhD thesis* (Division of Science, University of Otago)
 2018–2021 *University of Otago Doctoral Scholarship*
 2018 *Externally Funded Research Grant* (Max Planck Institute Plön)
 2015 *Summer Research Scholarship* (University of Auckland)
 2015 *PROMOS Travel Scholarship* (University of Greifswald)
 2014–2015 *Deutschlandstipendium* (Alfried Krupp von Bohlen und Halbach Foundation/Federal Government of Germany)

■ Additional Activities

- 2021 President of the Otago Computer Science Society (University of Otago)
 2019–2021 Member of the Postgraduate Committee (Department of Computer Science, University of Otago)
 2019–2021 Organising the annual Postgraduate Symposium (Department of Computer Science, University of Otago)
 2019–2021 Member of Student Council (Institute for Mathematics and Computer Science, University of Greifswald)

■ Talks

Conference Talks

- 2024 **Evolution 2024**, *Montreal (CA)*.
 Contributed talk: Instability in Phylogenetic Trees after Taxon Addition
 2023 **SMB meeting 2023**, *Columbus (OH, US)*.
 Invited minisymposium talk: Spaces of Discrete Time Trees
 2022 **Phylomania 2022**, *Hobart (AU)*.
 Contributed talk: Subtree Prune and Regraft on Ranked Trees
 2021 **Phylomania 2021** (Best Student Talk Award), *Online*.
 Contributed talk: Distances between Phylogenetic Time Trees
 2021 **NZ Phylogenomics Meeting**, *Akaroa (NZ)*.
 Contributed talk: The Space of Discrete Coalescent Trees
 2020 **NZ Phylogenomics Meeting**, *Waiheke (NZ)*.
 Contributed talk: Online Algorithms in Computational Biology
 2019 **NZ Phylogenomics Meeting**, *Napier (NZ)*.
 Contributed talk: The Ranked Nearest Neighbour Interchange Space of Phylogenetic Trees

Invited Seminar Talks

- 2025 **Department of Mathematics and Computer Science, University of Greifswald (GER).**
Using Deep Learning to Guide Phylogenetic Tree Search
- 2021 **Department of Mathematics, University of Otago (NZ).**
The Space of Discrete Coalescent Trees
- 2020 **Online Seminars on Algorithms and Complexity in Phylogenetics, Online.**
Computing the Ranked Nearest Neighbour Interchange Distance between Ranked Phylogenetic Trees
- 2019 **Max Planck Institute for Mathematics in the Science, Leipzig (GER).**
The Ranked Nearest Neighbour Interchange space of phylogenetic trees
- 2017 **Computational Evolution Group, ETH Zurich (CH).**
Discrete Time Trees

Poster Presentations

- 2024 **ICERM workshop “Algorithmic Advances and Implementation Challenges: Developing Practical Tools for Phylogenetic Inference”, Providence (RI, US).**
Poster: Instability in Phylogenetic Trees after Taxon Addition
- 2022 **Phylomania 2022, Hobart (AU).**
Poster: Subtree Prune and Regraft on Ranked Trees

Other

- 2025 **Association for Women in Mathematics, University of Texas (US).**
An Unexpected Journey
- 2022 **School of Mathematics and Statistics, University of Canterbury (NZ).**
How to Give a (Good) Talk
- 2020 **Postgraduate Symposium (1st place Best Presentation Award), University of Otago (NZ).**
The Complexity of Computing the RNNI Distance between Phylogenetic Trees
- 2020 **Seminar of Departments of Computer Science and Information Science, University of Otago (NZ).**
The Complexity of Computing Nearest Neighbour Interchange Distances between Ranked Phylogenetic Trees
- 2019 **Postgraduate Symposium (2nd place Best Presentation Award), University of Otago (NZ).**
Online Algorithms in Computational Biology

Publications

Collienne, L., Richman, H., Rich, DH., Barker, M., Jennings-Shaffer C., & Matsen IV, FA. (2026). Unifying phylogenetic traversal and deep learning to guide tree exploration *BioRxiv*. <https://doi.org/10.64898/2026.01.14.699358>

Collienne, L. & Matsen IV, FA. (2025). Tree rearrangement graphs admit paths of decreasing Robinson-Foulds distance *arXiv*. <http://arxiv.org/abs/2512.21397>

Collienne, L., Barker, M., Suchard, MA., & Matsen IV, FA. (2024). Phylogenetic tree instability after taxon addition: empirical frequency, predictability, and consequences for online inference. *Systematic Biology* syae059. <https://doi.org/10.1093/sysbio/syae059>

Collienne, L., Whidden, C. & Gavryushkin, A. (2024). Ranked Subtree Prune and Regraft. *Bulleting of Mathematical Biology* 86, 24. <https://doi.org/10.1007/s11538-023-01244-2>

Berling, L., **Collienne, L.** & Gavryushkin, A. (2024). Estimating the mean in the space of ranked phylogenetic trees. *Bioinformatics, Volume 40, Issue 8*. <https://doi.org/10.1093/bioinformatics/btae514>.

Collienne, L. (2021). Spaces of phylogenetic time trees (Thesis, Doctor of Philosophy). University of Otago. Retrieved from <http://hdl.handle.net/10523/12606>

Collienne, L., Elmes, K., Fischer, M., Bryant, D. & Gavryushkin, A. (2021). Discrete Coalescent Trees. *Journal of Mathematical Biology* 83.5, p. 60. issn: 1432-1416.

Collienne, L. & Gavryushkin, A. (2021). Computing nearest neighbour interchange distances between ranked phylogenetic trees. *Journal of Mathematical Biology* 82.1, p. 8. issn: 1432-1416.

Preprints

Collienne, L., Richman, H., Rich, DH., Barker, M., Jennings-Shaffer, C., & Matsen IV, FA. (2026). Unifying phylogenetic traversal and deep learning to guide tree exploration. *BioRxiv*. <https://www.biorxiv.org/content/10.64898/2026.01.14.699358v1>

Collienne, L. & Matsen IV, FA. (2025). Tree rearrangement graphs admit paths of decreasing Robinson-Foulds distance. *arXiv*. <http://arxiv.org/abs/2512.21397>

Bouckaert, R., **Collienne, L.** & Gavryushkin, A. (2022). Online Bayesian Analysis with BEAST2. *BioRxiv* <https://www.biorxiv.org/content/10.1101/2022.05.03.490538v1>.